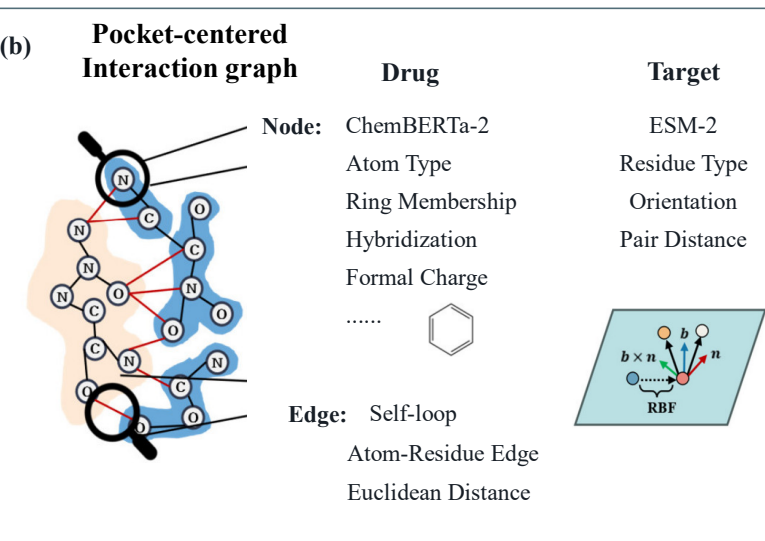




The diagram illustrates the proposed distillation framework. It consists of two main parts: a teacher model (top) and a student model (bottom), connected by a distillation loop.

Teacher Model (Top):

- Inputs $T_s(Z_s)$ and $T_g(Z_g)$ are processed by separate branches.
- $T_s(Z_s)$ is multiplied by weight w_s .
- $T_g(Z_g)$ is multiplied by weight w_g .
- The weighted outputs are summed ($\oplus$) to produce the unified representation Z^{uni} .
- Z^{uni} is passed through a "Distribution mapping" block to generate a "Distribution" (represented by a bar chart).

Student Model (Bottom):

- The student model takes Z^{multi} as input.
- Z^{multi} is passed through a "Distribution mapping" block to generate a "Distribution" (represented by a bar chart).

Distillation Process:

- A distillation loss is calculated between the teacher's and student's distributions, denoted as $\|p^{uni} - p^{multi}\|$.
- This loss is used for an "Update" step (indicated by a red dashed arrow) to refine the student model.